Assignment 4

Note: Q(A) & Q(B) are from Assignment 3.

(A). Write a C/C++ program to implement CPU scheduling. Data regarding process details need to be read from a file proc.dat

File format is as follows:

First line is number of processes N.

Each of next N lines contain ProcessID, Priority, Arrival time, followed by a sequence of CPU burst, IO burst. -1 indicates end of line. All these values are separated by comma. Following line indicates a process with id=2, priority 3, arrival time 4 and CPU bursts of 5, 4 and 7 interleaving with IO bursts of 2 and 3

2, 3, 4, 5, 2, 4, 3, 7, -1.

Last line contains value of quantum (by default and if this line is missing, quantum =1), Quantum is the interval at which timer interrupts.

Using this data, simulate following scheduling algorithms

a) FCFS
b) SJF (preemptive)
c) Priority (preemptive)
d) Round Robin

Your program should display progress of each process through a display with following symbols

C: running (CPU scheduled)

W: waiting for CPU

B; blocked for IO

R: running on IO.

For every process, history of its status has to be shown using above symbols.

(B). Write a program using C/C++ to simulate the FCFS, SJF (pre-emptive as well as non- pre-emptive approach). The scenario is: user may input n processes with respective CPU burst time and arrival time. System will ask the user to select the type of algorithm from the list mentioned above. System should display the waiting time for each process, average waiting time for the whole system, and final execution sequence.

(C). Write a program using C/C++ to simulate the priority scheduling (pre-emptive as well as non-preemptive approach) and RR, CPU scheduling algorithms. The scenario is: user may input n processes with respective CPU burst time and arrival time (also take the priority number in case of priority scheduling). System will ask the user to select the type of algorithm from the list mentioned above. System should display the waiting time for each process, average waiting time for the whole system, and final execution sequence.